# Ethics in NLP

Instructor: Lara J. Martin (she/they)

TA: Omkar Kulkarni (he)

https://laramartin.net/NLP-class/

# Learning Objectives

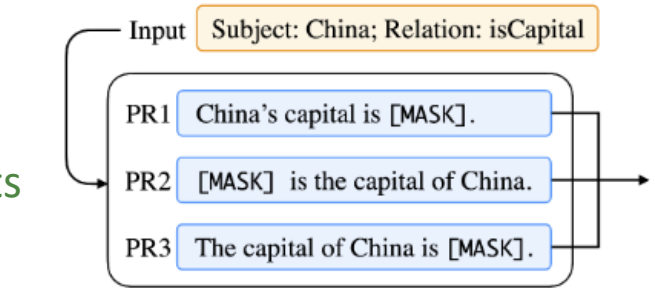
Identify ethical issues of LLMs/transformers from various lenses (social, environmental, legal, economic, etc.) by…

- Extracting them from the Stochastic Parrots paper
- Extending them with your own perspectives

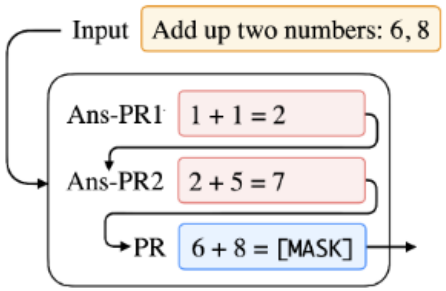
Determine how these issues apply to any LM

# Review: Multi-prompt "Learning"

Multiple unanswered prompts



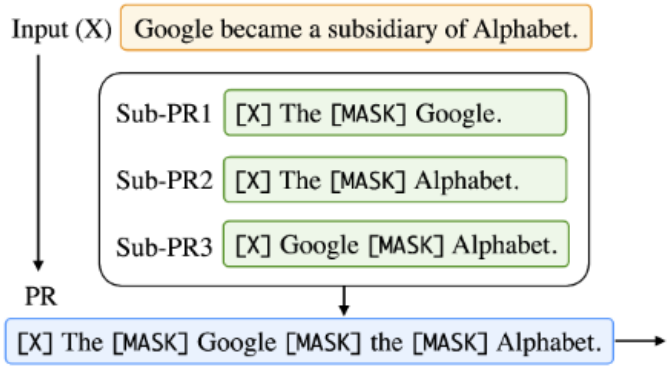

(a) Prompt Ensembling.



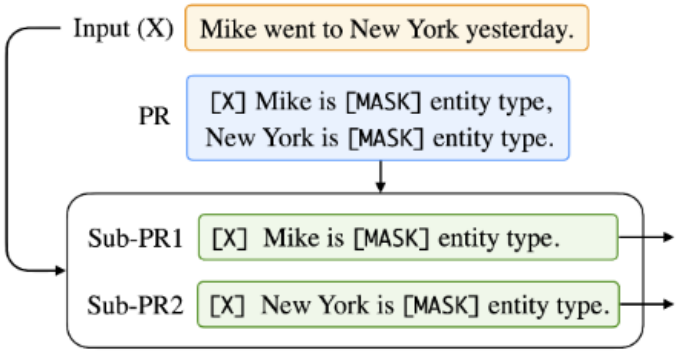

(b) Prompt Augmentation.

Demonstration learning

Merging subprompts




(c) Prompt Composition.




(d) Prompt Decomposition.

For multi-label predictions

"▭" for input text, "▭" for prompt, "▭" for answered prompt. "▭" for sub-prompt.



Pengfei Liu, Weizhe Yuan, Jinlan Fu, Zhengbao Jiang, Hiroaki Hayashi, and Graham Neubig. 2023. Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing. ACM Comput. Surv. 55, 9, Article 195 (September 2023), 35 pages. https://doi.org/10.1145/3560815

# Review: Self-Criticism

LLM “ruminates” on its output to try to come up with better output

(Along with chain-of-thought) Precursor to reasoning models that are finetuned to do this automatically

```
Question: Who was the third president of the United States?
Here are some brainstormed ideas: James Monroe
Thomas Jefferson
John Adams
Thomas Jefferson
George Washington
Possible Answer: James Monroe
Is the possible answer:
 (A) True
 (B) False
The possible answer is:
```



Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, *et al.* 2022. Language Models (Mostly) Know What They Know. arXiv: 2207.05221

# Review: Meta-Prompting

Prompting to generate prompts

Meta-prompts…

- “encode what the LLM should “do” given different descriptions of the same task” [1]
- “will return the relevant outputs for an arbitrary task, provided that the task description is provided as an input” [1]



[1] Adrian de Wynter, Xun Wang, Qilong Gu, Si-Qing Chen. 2023. On Meta-Prompting. arXiv: 2312.06562

# Stochastic Parrots

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 610–623. https://doi.org/10.1145/3442188.3445922

# Ethical Issues

1. Environmental
2. Financial
3. Diversity
4. Static Data
5. Bias
6. Accountability
7. Lack of Understanding
8. Subjective Coherence
9. Harms

+ 10. Mitigation Strategies

# 1) Environmental

# 2) Financial

# 3) Diversity

# 4) Static Data

# 5) Bias

# 6) Accountability

# 7) Lack of Understanding

# 8) Subjective Coherence

# 9) Harms

# 10) Mitigation Strategies

# Issues of all LMs

How many of these are relevant to smaller LMs?